BHAVYA KANSAL

AI/ML Engineer | Deep Learning Researcher | Computer Vision & Generative AI

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PROFESSIONAL SUMMARY

AI/ML Engineer and Deep Learning Researcher pursuing B.Tech (AI & Data Science) at Thapar Institute of Engineering and Technology, with a Diploma in Computer Science Engineering and a 6-month AI/ML internship at IIT Ropar. Skilled in end-to-end AI product development — from model design and training to full-stack cloud deployment — with 3 production-grade systems shipped live across healthcare diagnostics, computer-vision restoration, and multimodal AI assistants. Seeking research collaborations, internships, and AI/ML roles where I can build and ship high-impact, production-scale AI systems.

EDUCATION

Thapar Institute of Engineering and Technology (TIET) 2026 – Present

B.Tech in Data Science and Artificial Intelligence — Patiala, Punjab

Thapar Polytechnic College Aug 2023 – Jun 2026

Diploma in Computer Science Engineering — Patiala, Punjab

EXPERIENCE

AI/ML Intern — IIT Ropar Jan 2026 – Jul 2026

Ropar, Punjab

- Completed a 6-month AI/ML internship spanning deep learning, computer vision, NLP, and multimodal AI.
- Delivered 3 production-grade AI projects as core deliverables, including the multi-disease Healthy AI platform.

PROJECTS

Healthy AI — Multi-Disease Medical AI Platform

- Built a production-ready medical AI platform spanning heart disease prediction, diabetes risk assessment, and brain tumor MRI classification.
- Achieved 91.62% accuracy on 4-class tumor classification via a fine-tuned deep CNN; deployed on Render with a dark-themed Flask interface.

GitHub: github.com/BhavyaKansal20/healthy-ai | Live: healthyai-dlog.onrender.com

ChromaCrystal UHD — AI Photo Colorization & Restoration Platform

- Engineered a latency-guaranteed FastAPI pipeline (DDColor + GFPGAN + Real-ESRGAN) serving 20+ concurrent users under a strict 20-second SLA on a 2-core, 16GB free-tier server.
- Cut per-image latency 83% (15s → 2.5s) with dynamic load-shedding, and model-load time 30s → 0s via global caching; mutex-isolated inference eliminated OOM crashes entirely.
- Built a Next.js 15 (TypeScript) frontend at a consistent 60 FPS with WebSocket-based real-time job tracking; deployed on Hugging Face Spaces.

GitHub: github.com/BhavyaKansal20/ChromaCrystal_UHD | Live: bhavyakansal20-chromacrystal-uhd.hf.space

Aagni AI — Sovereign Multimodal AI Assistant

- Built "Aagni," a multimodal AI assistant delivering context-aware conversational intelligence across Telegram and web.
- Engineered a Next.js + PostgreSQL/Prisma backend integrating Sarvam AI's multimodal models with Google/GitHub OAuth; successfully tested with 20+ users pre-commercial launch.

Telegram: t.me/AagniAIbot | Live: aagni-ai.veercel.app

TECHNICAL SKILLS

Languages: Python (Expert), HTML/CSS/JS, Java (Intermediate)

ML / DL Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, OpenCV, MediaPipe

Deep Learning: CNN, MobileNetV2, EfficientNet-B0, LSTM, BiLSTM, GAN (Basic), Transformer architecture

NLP & GenAI: NLP pipelines, sequence modeling, LLM fine-tuning, RAG, ReAct agent patterns

Data & Analysis: Pandas, NumPy, Matplotlib, Seaborn, EDA, data preprocessing pipelines

Deployment & Tools: Hugging Face Spaces, Render, Vercel, Docker, Git, VS Code, Colab, Kaggle

RESEARCH & ACHIEVEMENTS

- **AI in Healthcare:** Researching diabetic retinopathy grading via fundus image deep learning and explainable AI (XAI).
- **Generative AI & Multimodal Systems:** Researching PEFT/LoRA fine-tuning, RAG pipelines, and cross-modal fusion.

LEADERSHIP & POSITIONS OF RESPONSIBILITY

General Secretary — Code Metrics Society, TIET 2026 – Present

- Selected to lead a TIET society focused on AI, software engineering, and data-analytics research; led interview-based recruitment of the founding executive team.
- Represented the society at the TIET society fair, engaging 500+ students through interactive AI/ML demo activities.